

Solve the following system:

$$\begin{cases} e^{2x} + 2\ln y = 5 \\ e^{4x} + \ln y = 4 \end{cases}$$

Solve:

$$\ln y = 4 - e^{4x} \Rightarrow e^{2x} + 2(4 - e^{4x}) = 5$$

$$e^{2x} + 8 - 2e^{4x} - 5 = 0$$

$$-2e^{4x} + e^{2x} + 3 = 0$$

$$2e^{4x} - e^{2x} - 3 = 0$$

$$\text{Let } t = e^{2x}$$

$$\Rightarrow 2t^2 - t - 3 = 0$$

$$t_1 = 1$$

accepted

$$1 > 0$$

$$\Rightarrow e^{2x} = 1$$

$$\text{so: } e^0 = 1$$

$$\Rightarrow \boxed{x = 0}$$

$$\text{and } t_2 = -\frac{3}{2}$$

rejected

$$e > 0$$

$$\text{then: } e^{4(0)} + \ln y = 4$$

$$\ln y = 3$$
$$\boxed{y = e^3} \Rightarrow \int: \{[0, e^3]\}$$

Solve the following system:

$$e^{2x} + 2\ln y = 6$$

$$e^{4x} + \ln y = 3$$

$$\ln y = 3 - e^{4x} \Rightarrow e^{2x} + 2(3 - e^{4x}) = 6$$

$$e^{2x} + 6 - 2e^{4x} - 6 = 0$$

$$e^{2x} - 2e^{4x} = 0$$

$$\text{Let } t = e^{2x}$$

$$-2t^2 + t = 0$$

$$2t^2 - t = 0$$

$$t_1 = \frac{1}{2}$$

$$e^{2x} = \frac{1}{2}$$

~~$$e^{2x} = \frac{1}{2}$$~~

~~$$\frac{1}{2} + 2\ln y = 6$$~~

$$2\ln y = 6 - \frac{1}{2}$$

$$\ln y = \frac{11}{4} \Rightarrow y = e^{11/4}$$

$$t_2 = 0$$

$$e^{2x} = 0 \Rightarrow \text{rejected}$$

$$e > 0$$

~~$$e^{2x} = 0$$~~

$$e^{2x} = \frac{1}{2}$$

$$2x = \ln\left(\frac{1}{2}\right)$$

$$x = -\frac{\ln 2}{2}$$

$$S: \left\{ \left(-\frac{\ln 2}{2}, e^{11/4} \right) \right\}$$